CL246 Heat Transfer

Analysis of Heat Transfer for a Martian Structure



GROUP 24

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1 Introduction

Mars is our nearest potentially habitable planet which shows lots of promise for the future. Despite all the challenges ahead, humanity plans to colonize the planet.

Figure 1: The Future



1.1 User Story

Structures where human can sustain themselves on the harsh Martian surface are important if we wish to colonize the red planet. Since atmospheric conditions on Mars are vastly different from that on Earth, it is important to create new and innovative designs but at the same time it also necessary that we first look at designs used in harsh climates at Earth and analyse how these would perform on Mars. We have started with a simple shape prevalent in places with harsher climates and performed analysis on it using performance parameters such as heat losses and energy requirements of the structure. This serves as a building block to understand how the design performs and where are it's limitations, which can then be improved upon in pursuit of building an ideal structure for Mars.

1.2 Existing Designs

While researching designs found in the colder regions of earth, we stumbled upon a particular structure called as **Dome Home**. This structure has proven to be extremely energy efficient due to its design and lower costs of building. They are also more resistant and sturdy during natural disasters such as earthquakes and would perform relatively better than more conventional shaped houses in dust storms that are commonplace on mars.

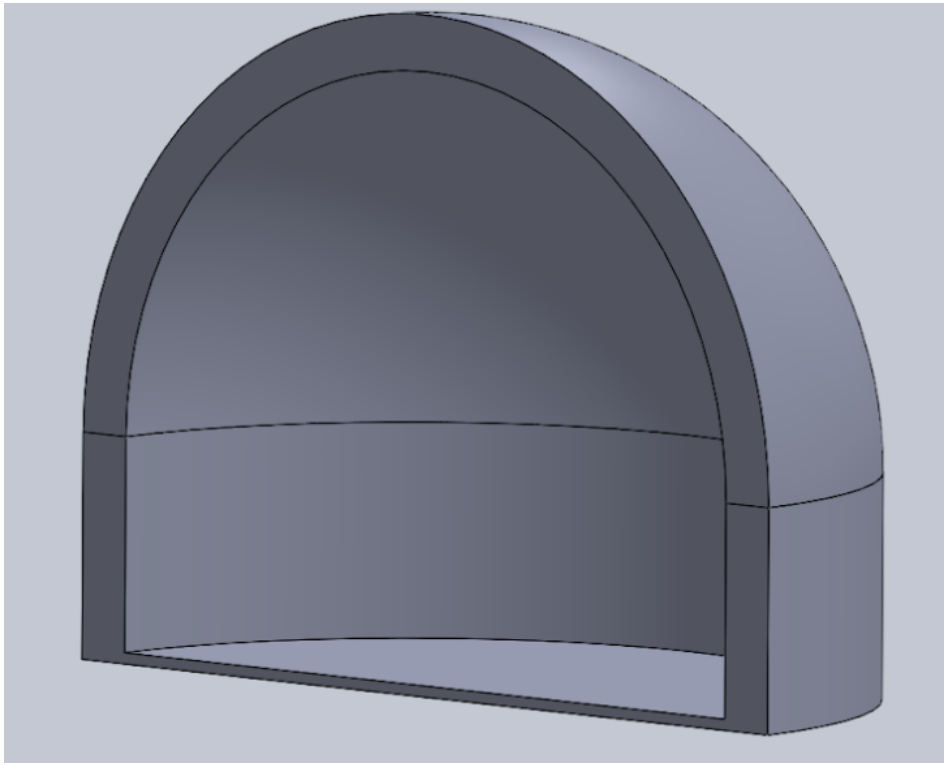
Figure 2: Dome Home



1.3 Technical Problem

Consider the dome shaped structured (hemisphere above cylinder) house situated in Mars. Average temperature is maintained by a constant heat source inside the structure.

Figure 3: Simplified Design



We have the following quantities available to us (or we have made suitable assumptions) :-

- Fixed inner wall temperature: 295.15 K
- Atmospheric temperature of mars: 213.15 K
- Outer diameter of structure: 5 m
- Thickness of wall: 0.5 m
- Thermal Conductivity of wall (k) = 1.2 $\mathrm{W\,m^{-1}\,K^{-1}}$

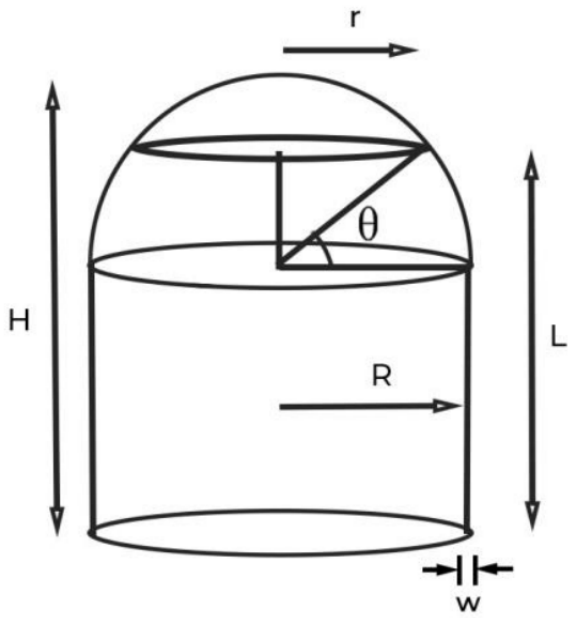
We calculate the following using appropriate relations in this report :-

- Convective heat transfer coefficient for heat exchange between outer wall of the structure and atmosphere
- Temperature of outer wall of structure.
- Total heat loss from the structure.
- Average cost of heating per day

2 Schematic and Control Volume

Structure consists of cylindrical wall structure along with hemispherical dome as top.

Figure 4: Complete Structure



Where :-

- R (inner radius of structure) = 2 m
- H-R (height of cylindrical part) = 3 m
- w (thickness of wall) = 0.5 m
- θ = Angle made by any angular disc at a high L and radius r

2.1 Control Volume

Control volume consists of the cylindrical control volume and hemispherical control volume that covers entire structure.

Figure 5: Complete Control Volume

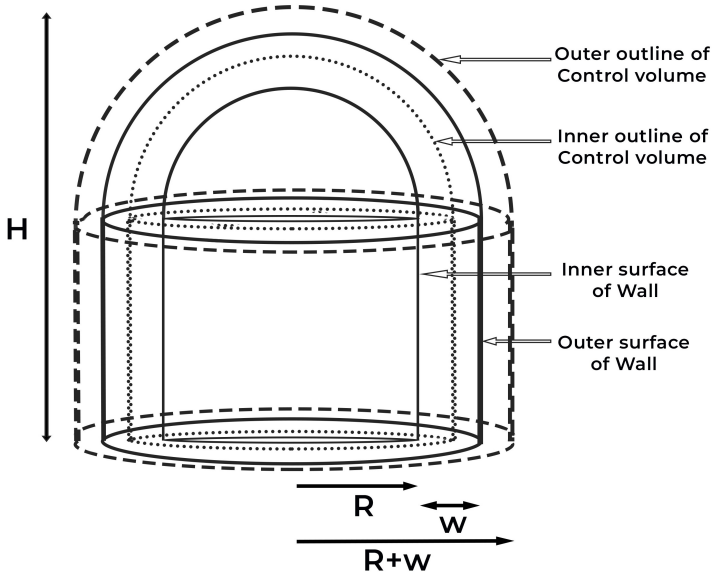
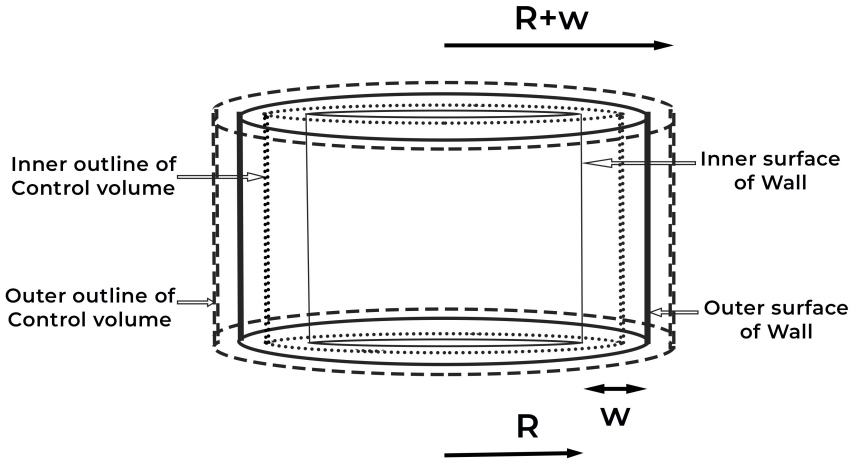


Figure 6: Individual CV

(a) Cyclendrical Control Volume



(b) Hemispherical Control Volume

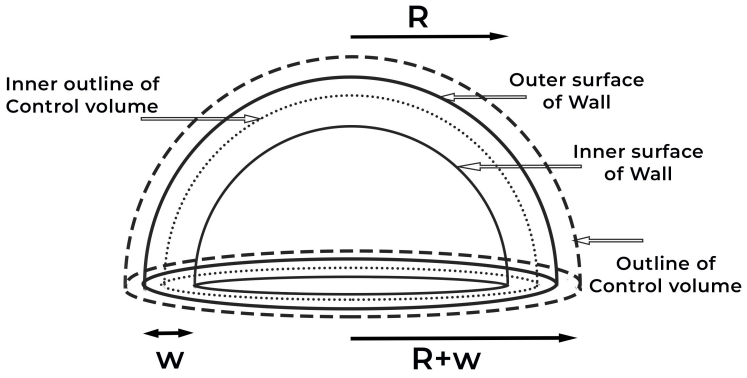
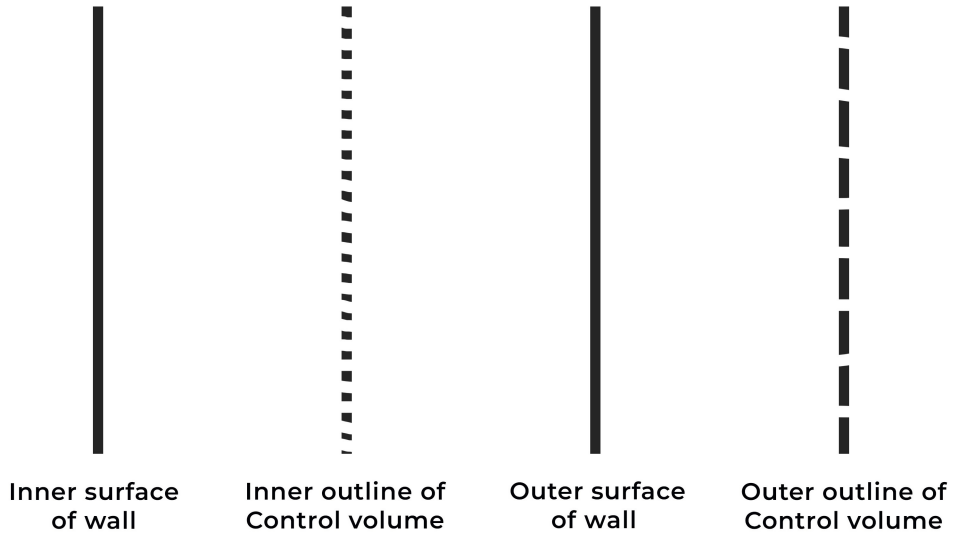


Figure 7: Planar Diagram



3 Solutions and Methodology

3.1 Heat Transfer Coefficient

In order to calculate convective heat losses, we have to calculate heat transfer coefficients first.

3.1.1 Properties and Estimates

We are considering the ambient temperature (Ta) as $-60°$ C which is the average over a summer day on the equator. We have kept the Inner wall temperature fixed whereas outer wall temperature can vary. We have extrapolated the values of properties for CO_2 which were given in the Appendix (Heat Transfer Textbook - Incropera). The value of the Prandtl Number is 0.7627. Average wind velocity over the surface of Mars is taken to be 5 m/s ignoring special cases such as dust storms. These values are used to calculate the Heat Transfer coefficients for the cylinder and dome structures. Finally, the paper has also approximated the hemisphere (referred to as dome) as a sphere. This can be done since we are only concerned with the top part and we are setting up a 1-D heat transfer problem (in the radial direction in case of the hemisphere). Since the direction or the solid angle cast by the object in question does not factor into the problem, making such an assumption is justified.

3.1.2 Free Convection for Cylinder

We approximated the cylindrical geometry as a vertical plate and calculated the Rayleigh Number using the formula:

$$Ra_L = \frac{g\beta L^3 \left(T_{s,cylinder} - T_\infty\right)}{\nu\alpha}$$

The value obtained ($\mathrm{Ra} = 7.2760e+11$) which is within limits to use the correlation provided by Churchill and Chu to get the average Nusselts number:

$$\overline{Nu}_{CFree} = \left\{ 0.825 + \frac{0.387 Ra_L^{\frac{1}{6}}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{\frac{9}{16}}\right]^{\frac{8}{27}}} \right\}^2$$

Finally we get the value of the average heat transfer coefficient for free convection using:

$$h_{CFree} = \frac{Nu_{CFree} k}{L}$$

The value comes out to be 4.0466 W/m^2K

3.1.3 Forced Convection for Cylinder

For forced convection around the cylinder, we have taken the average velocity on the surface of Mars to be 5 m/s. This is used to calculate the Reynold's Number for the cylinder using the outer diameter as the length scale:

$$Re_{CFor} = \frac{VD}{\nu}$$

The value of Re comes out to be $5.0766e+06$. The relation for the average Nusselts number given by Churchill and Bernstein is used (recommended for $\mathrm{RePr} \geq 0.2$):

$$\overline{Nu}_{CFor} = 0.3 + \frac{0.62\,\mathrm{Re}_D^{1/2}\,\mathrm{Pr}^{1/3}}{\left[1 + (0.4/Pr)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{Re_D}{282,000}\right)^{5/8}\right]^{4/5}$$

Finally the average heat transfer coefficient for forced convection around the cylinder is calculated:

$$h_{CFor} = \frac{Nu_{CFor} k}{L}$$

It's value comes out to be 12.9843 W/m^2K

3.1.4 Free Convection for Dome

For natural convection around the dome, we recalculate the atmospheric properties as our film temperature changes in the two cases (cylinder and dome). We calculate the Rayleigh's number for the dome using the formula:

$$Ra_{DFree} = \frac{g\beta\ D^3\left(T_{s,dome} - T_\infty\right)}{\nu\alpha}$$

The value for Ra comes out to be 7.0687e+12. We use the correlation for average Nusselts number given by Churchill for free convection around a sphere even though the recommend Rayleigh's number (Ra $\leq 10^{11}$) is less than our calculated number since no other suitable replacement is available:

$$\overline{Nu}_{DFree} = 2 + \frac{0.589 Ra_D^{1/4}}{\left[1 + (0.469/\ \mathrm{Pr})^{9/16}\right]^{4/9}}$$

Then we calculate the average HT coefficient using:

$$h_{DFree} = \frac{Nu_{DFree} k}{L}$$

The value for h we get is 1.8722 W/m^2K

3.1.5 Forced Convection for Dome

For forced convection around the dome (sphere), we start by calculating the Reynold's number assuming the lenght scale to be the outer diameter of the dome:

$$Re_{DFor} = \frac{VD}{\nu}$$

The value obtained is 5.4301e+06. The correlation for the average Nusselts number given by Whitaker is used. All properties are recalculated at ambient temperature with the exception of the dynamic viscosity (μ) which is calculated at the surface temperature of the outer wall:

$$\overline{Nu}_{DFor} = 2 + \left(0.4 Re_D^{1/2} + 0.06 Re_D^{2/3}\right) \mathrm{Pr}^{0.4} \left(\frac{\mu}{\mu_s}\right)^{1/4}$$

Then we calculate the average HT coeeficient using:

$$h_{DFor} = \frac{Nu_{DFor} \cdot k}{D}$$

The value of h comes out to be 6.1843 W/m^2K

3.1.6 Net Heat Transfer Coefficient

We calculate the net heat transfer coefficient (combined HT coefficient for free and forced convection) using the following relations:

$$h_C = (h_{CFree}^3 + h_{CFor}^3)^{\frac{1}{3}}$$

The value comes out to be 13.1227 W/m^2K for the cylinder and

$$h_D = (h_{DFree}^3 + h_{DFor}^3)^{\frac{1}{3}}$$

for the dome it comes out to be 6.2396 W/m^2K

3.2 Equations

3.2.1 Radiation

The variation of the solar radiation on the Martian surface is governed by three factors:
(1) the Mars-Sun distance,
(2) solar zenith angle, and
(3) by the opacity of the Martian atmosphere.
The global solar radiation is composed of the direct beam and diffuse components. The direct beam irradiance, G_b, on the Martian surface normal to the solar rays is related by Beer's law to the optical depth, of the intervening atmospheric haze:

$$G_b = G_{ob} exp[\tau m(z)]$$

where τ is optical depth,
where m(z) is the air mass determined by the zenith angle z

$$m(z) = \frac{1}{cosz}$$

The solar irradiance components, on a horizontal Martian surface, are related by:

$$G_h = G_{bh} + G_{dh}$$

where
G_h global irradiance on a horizontal surface
G_{bh} direct beam irradiance on a horizontal surface
G_{dh} diffuse irradiance on a horizontal surface

$$G_h = G_{ob} cosz \frac{f(z,\tau)}{0.9}$$

$$G_{bh} = G_{ob} cosz exp(\frac{-\tau}{cosz})$$

Following are the tables which gives the values of beam isolation and diffuse isolation:-

TABLE V. - HOURLY AND DAILY BEAM INSOLATION ON A HORIZONTAL SURFACE AT MARS SURFACE
[VL1: ϕ = 22.3°N]

Hourly beam insolation* I_{bh} (Whr/m^2-hr) for hours ending at:									Daily beam insolation*, H_{bh}, Whr/m^{2}day	Daylight hours* T_d, hr	Daily mean beam irradiance, W/m^2
Day-L_s	τ	13:00	14:00	15:00	16:00	17:00	18:00	19:00			
69°	0.65	252	230	186	128	67	20	3	1768	13.34	133
120°	.40	352	322	265	190	103	33	2	2534	13.24	191
153°	.50	345	310	244	163	77	15	--	2308	12.62	183
249°	1.40	69	50	26	10	2	--	--	314	10.66	29
299°	3.25	3	2	1	---	--	--	--	12	10.75	1

*Multiply by 24.65/24 to obtain actual (terrestrial) Whr/m^2 or hours

TABLE VI. - HOURLY AND DAILY DIFFUSE INSOLATION ON A HORIZONTAL SURFACE AT MARS SURFACE
[VL1: = 22.3°N]

Hourly diffuse insolation* I_{dh} (Whr/m^2-hr) for hours ending at:									Daily diffuse insolation,* H_{dh}, Whr/m^{2}day	Daylight hours* T_d, hr	Daily mean diffuse irradiance, W/m^2
Day-L_s	τ	13:00	14:00	15:00	16:00	17:00	18:00	19:00			
69°	0.65	168	160	152	135	103	58	10	1572	13.34	118
120°	.40	125	124	122	116	98	65	13	1326	13.24	100
153°	.50	163	161	155	139	108	58	3	1574	12.62	125
249°	1.40	238	220	178	112	43	2	--	1586	10.66	149
299°	3.25	167	147	106	61	24	1	--	1012	10.75	94

*Multiply by 24.65/24 to obtain actual (terrestrial) Whr/m^2 or hours

where,

$$I_{bh} = \frac{12^*}{\pi} G_{ob} \int_{\omega_1}^{\omega_2} (\sin\phi \sin\delta + \cos\phi \cos\delta \cos\omega) \exp\left[-\tau/(\sin\phi \sin\delta + \cos\phi \cos\delta \cos\omega)\right] d\omega$$

ϕ latitude

δ declination angle

ω hour angle measured from the true noon westward

Maximum radiation for a given time in day is obtained at 153° during Total Radiation(Max) = $\frac{2308+1574}{24} \approx 160\ \frac{W}{m^2}$

3.2.2 Conduction

We have considered 1D stead state conduction within the wall. Heat transfers from the inner wall to the outer wall. The material of the wall is taken to be silicon dioxide (SiO_2) which is the primary component of glass. It has a thermal conductivity of 1.2 W/mK, glass aluminosilicate is a form of glass closest to SiO_2 with thermal conductivity 1.225 W/mK (used for ansys simulations).

The equation for both the dome and cylinder looks like:

$$q_{cond} = kw\delta T$$

where k is the thermal conductivity, w is the width of the wall and δT is the difference between the temperature of the inner wall and the outer wall.

3.2.3 Convection

Since we have already calculated the heat transfer coefficients for the dome as well as the cylinder, we can now calculate the heat loss through convection to the atmosphere from the outer wall of the structure:

$$q_{conv} = h(\text{Surface Area})\delta T$$

where h is the convective heat transfer coefficient and δT is the difference between the surface temperature of the outer wall and the ambient temperature.

3.2.4 Outer wall temperature

We do not know the outer wall surface temperature, to calculate the same under steady state conditions, we make use of the equations listed uptil now and balance the heat gained by the outer wall and the heat lost by the outer wall to the atmosphere:

$$q_{cond} + \text{solar radiation} = q_{conv}$$

Subsituting the equations defined earlier into this equation we get:

$$kw(T_i - T_s) + 160A = hA(T_s - T_a)$$

where A is the surface area, k is the thermal conductivity, h is the HT coefficient, T_s is the outer wall surface temperature, T_i is the inner wall surface temperature and T_a is the ambient temperature. Taking all T_s to the LHS, we get:

$$T_s = \frac{kwT_i + hAT_a + 160A}{hA + kw}$$

We solve these set of equations iteratively until the difference between subsequent surface temperature becomes negligible since all correlations used earlier also directly or indirectly depended on this value. Once the difference between subsequent temperatures obtained is less that 0.01, we break the loop. The value obtained of the outer wall temperature for the cylinder is $-47.7398°$ C and for the dome it's $-34.2198°$ C.

3.2.5 Heat loss

Finally we calculate the net heat loss from the structure which is nothing but the heat exchanged with the atmosphere through convection. The value of the heat loss comes out to be 160.888 W/m^2 for the cylinder and 160.8590 W/m^2 for the dome.

3.3 Steady State Temperature Profile

The temperature difference in the boundary layer is more in case Dome as compared to that of Cylindrical region. Presence of radiation also increases this difference in both the cases. The profile for Dome and Cylinder are as follows:-

Figure 8: 1D Profile of Dome

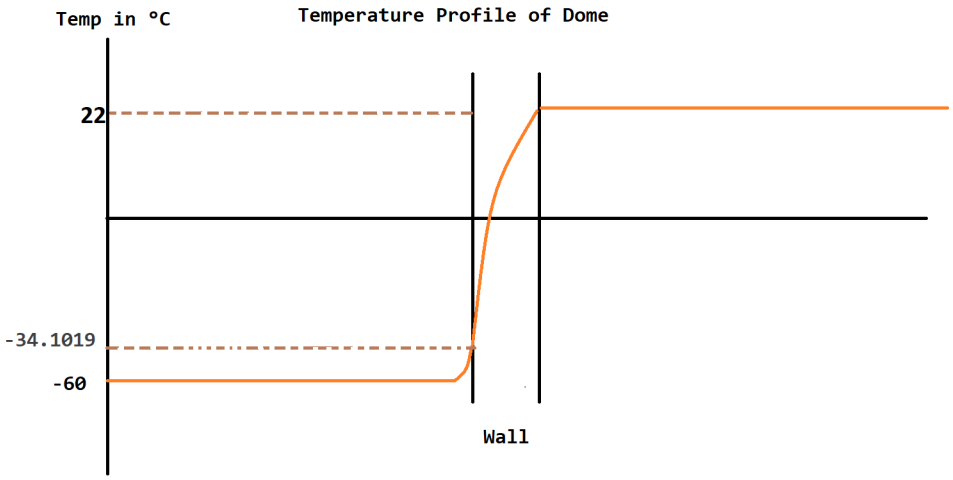
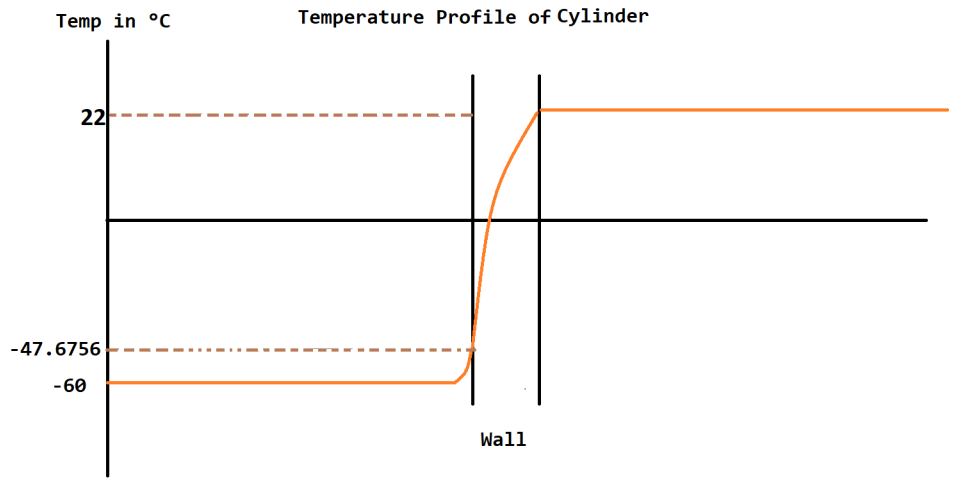


Figure 9: 1D Profile of Cylinder



The profile of conduction in the wall is not linear due to presence of radiation. The material used allows transmission of heat through it making the profile non-linear.

3.4 Ansys Simulation

3.4.1 Comparison of Results

The simulation solutions are highlighted in the following section. The outer temperatures of the wall come out to be $-20.656°$ C for the dome and $-30.431°$ C for the cylinder. The heat flux going out through the outer surface of the dome and cylinder are:

$$q_{dome} = 136.56 W/m^2$$
$$q_{cylinder} = 149.59 W/m^2$$

3.4.2 Simulation Images

Figure 10: Heat flux through dome

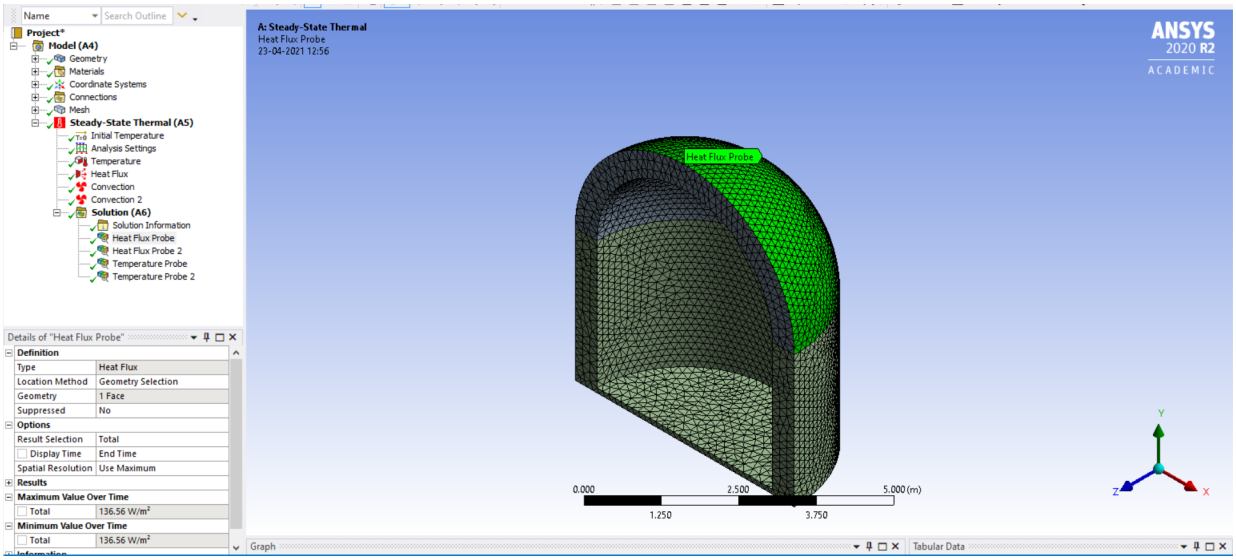


Figure 11: Heat flux through cylinder

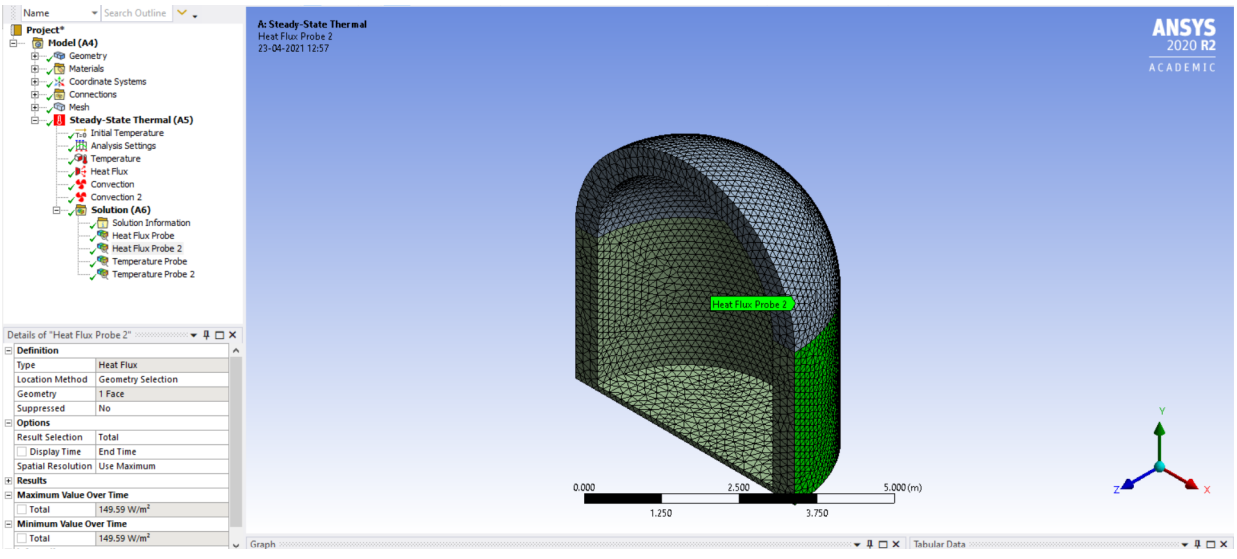


Figure 12: Outer surface temperature of Dome

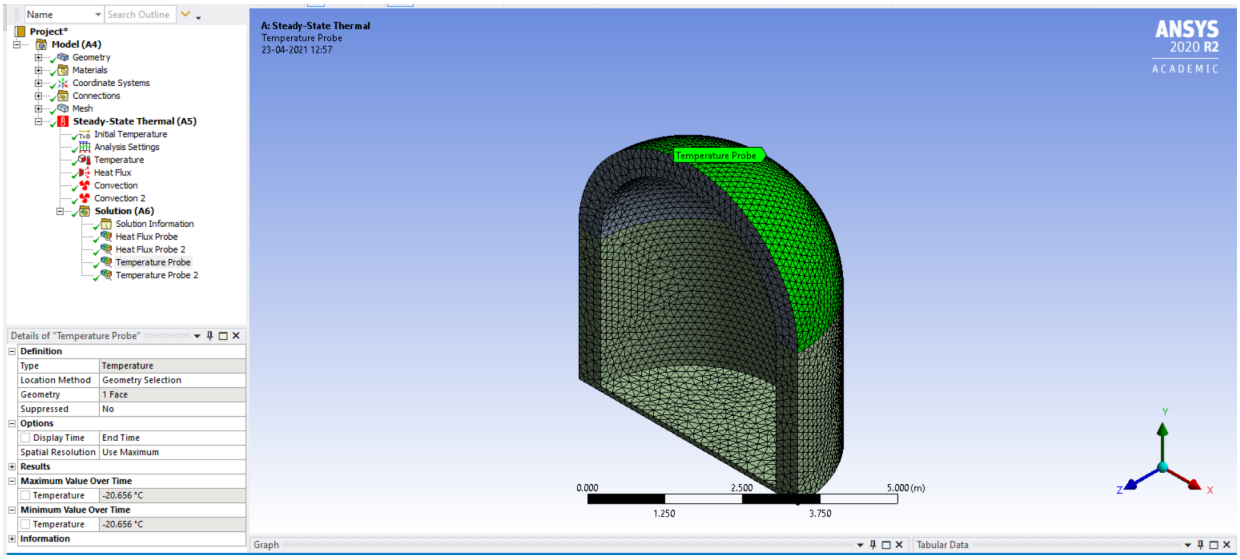
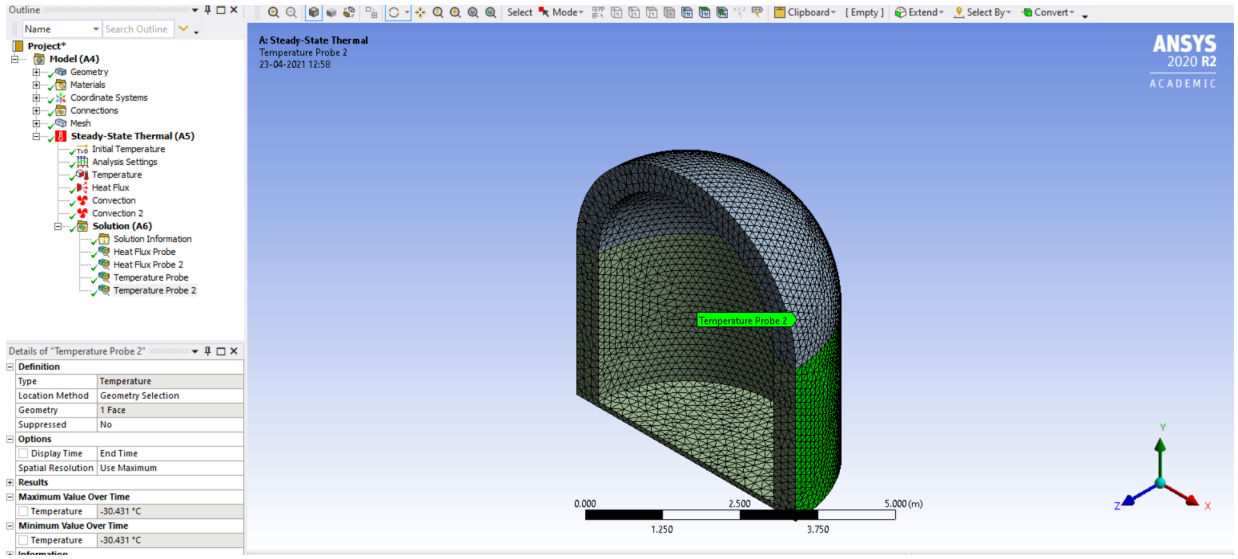


Figure 13: Outer surface temperature of Cylinder



3.4.3 Reasons for discrepancy between results

- ANSYS took thermal expansion of the body into consideration which we didn't. Since cylinder expands radially but dome expands both ways, deviation from simulation is higher for dome.
- ANSYS solved a transient equation assuming inner surface temperature -60°C in the beginning while we solved for steady state.
- We extrapolated the data given in the book for all required properties
- We used the best possible correlations available but those can still give rise to some error

3.5 Cost Calculations

Since we now know the heat lost to the surrounding, under the assumption that the heat generated by the heaters inside is equal to the heat lost (i.e $Q_{lost} = Q_{generated}$, we can calculate the cost to maintain the required temperature.

Since we wanted to study a standard Earth house, we took electricity costs present here. According to Statista's Repoort, average cost of 1 kW h was ₹6.09 in the year 2019.

Calculations are as follows:-

$$Q_{lossDome} = q_{lossDome} * 0.5 * \pi * D^2 = 6316.85 \mathrm{J\,s^{-1}}$$

$$Q_{lossCyc} = q_{lossCyc} * \pi * L * D = 7581.63 \mathrm{J\,s^{-1}}$$

$$Q_{generated} = Q_{lossTotal} = Q_{lossDome} + Q_{lossCyc} = 13898.48 \mathrm{J\,s^{-1}}$$

$$T_{Day} = 60\mathrm{s} * 60(min) * 24(hrs) = 86400\mathrm{s}$$

$$1\mathrm{kW\,h} = 3.6 * 10^6 \mathrm{J}$$

$$\text{Total Cost} = ₹2031.4 \text{ per Day}$$

4 Conclusion

We conclude this report with the following results:
The heat lost to the surroundings from the dome part of the enclosure is equal to 6316.85 Joule per second The heat lost to the surroundings from the cylindrical part of the enclosure is equal to 7581.63 Joule per second Heat generated should be equal to the total heat lost to maintain the temperature within the enclosure. Therefore heat generated should be equal to 13898.48 Joule per second Total cost to maintain the temperature of the enclosure is around Rs 2031.4 per day The outer air temperature on Mars is approximated to be -60 degree Celsius We conclude that our structure has performed well to sustain life on Mars. The temperature is maintained well within our enclosure. The calculations of the heat transfer rate are close to the values obtained in the Ansys simulations of the enclosure in a Martian atmosphere.

At the end of this report we have attached the code used to calculate the outer surface temperature and heat loss from the structure.

```
% All calculations are done assuming the temperature to be the average
% during a summer day at the equator. The process is iteratively repeated
% till the correct outer surface temperature is reached. The material of
% the wall is assumed to be made of Silicon Dioxide primarily(primary
% component of glass).
% Once the actual outer surface temperature is determined, the net heat loss
% from the wall to the surroundings is expressed as qloss [W].
% Atmosphere is assumed to be primarily carbon dioxide and the values for
% the constants needed in different equations is extrapolated from the
% appendix of the book.
% Average velocity on surface of mars is considered to be 5m/s

% constants
Ta = -60; %ambient air temperature
g = 3.721; %gravity on mars
L = 3; %length of cylinder
V = 5; %average velocity
w = 0.5; %thickness of wall
D = 5; %outer diameter of structure
kwall = 1.2; %thermal conductivity of silicon dioxide
Ti = 22; %Fixed inner wall temperature
Tsactualcylinder = 0; %variable initialization
Tscylinder = 1; %variable intialization
Tsactualdome = 0; %variable initialization
Tsdome = 1;

while (abs(Tsactualcylinder - Tscylinder)>0.01 && abs(Tsactualdome - Tsdome)>0.01)

% Surface temperature and dependents of temperature
Tscylinder = Tsactualcylinder;
Tsdome = Tsactualdome;

Tfcylinder = (Ta + Tscylinder)/2;
nu = (7.36 + ((8.4-7.36)/20)*(Tfcylinder+7))*10^(-6); %extrapolated value for CO2
alpha = (9.63 + ((11-9.63)/20)*(Tfcylinder+7))*10^(-6); %extrapolated value for CO2
k = (15.2 + ((16.55-15.2)/20)*(Tfcylinder+7))*10^(-3); %extrapolated value for CO2
Pr = (0.765 + ((0.766-0.765)/20)*(Tfcylinder+7)); %extrapolated value for CO2

% Natural around cylinder

%Cylinder is assumed to be a vertical plate for free convection
RaL = (g/abs(Tfcylinder))*(L^3)*((Tscylinder - Ta)/(nu*alpha)); %Rayleigh Number
%Since Ra comes to be greater than 10^9, Churchill and Chu correlation is
%used for Nu
Nu = (0.825 + ((0.387*(RaL^(1/6)))/(1 + (0.492/Pr)^(9/16))^(8/27)))^2;
hcyl = Nu*k/L;

% Forced around cylinder

ReD = V*D/nu;
%For this external flow around a cylinder is used and the correlation
%recommended by Churchill and Bernstein is used. It is valid for RePr >=
%0.2 which is true in this case.
```

```
NuDforced = 0.3 + ((0.62*(ReD^0.5)*(Pr^(1/3)))/(1 + (0.4/Pr)^(2/3))^0.25)*(1 + (ReD/282000)^(5/
hcylforced = NuDforced*k/D;

Tfdome = (Ta + Tsdome)/2;
nu = (7.36 + ((8.4-7.36)/20)*(Tfdome+7))*10^(-6); %extrapolated value for CO2
alpha = (9.63 + ((11-9.63)/20)*(Tfdome+7))*10^(-6); %extrapolated value for CO2
k = (15.2 + ((16.55-15.2)/20)*(Tfdome+7))*10^(-3); %extrapolated value for CO2
Pr = (0.765 + ((0.766-0.765)/20)*(Tfdome+7)); %extrapolated value for CO2

% Natural around dome

%approximated as a sphere
RaD = (g/abs(Tfdome))*(D^3)*((Tsdome - Ta)/(nu*alpha));
%again Turbulent since Ra is around 10^12 but since no other relation is
%know, it is assumed to be good enough for the 10^11 relation, i.e the
%correlation for spheres in fluids by Churchill is used
NuD = 2 + ((0.589*(RaD^0.25))/(1 + (0.469/Pr)^(9/16))^(4/9));
hdome = NuD*k/D;

% Forced around dome

%finally for forced around a dome, the dome is approximated as a sphere and
%the correlation by Whitaker is used to calculate h after calculating all
%the properties at ambient temperature.
nudomeforced = (7.36 + ((8.4-7.36)/20)*(Ta+7))*10^(-6);
alphadomeforced = (9.63 + ((11-9.63)/20)*(Ta+7))*10^(-6);
kdomeforced = (15.2 + ((16.55-15.2)/20)*(Ta+7))*10^(-3);
Prdomeforced = (0.765 + ((0.766-0.765)/20)*(7+Ta));
ReDomeforced = V*D/nudomeforced;
muTa = (140 + ((149-140)/20)*(Ta+7))*10^(-7);
muS = (140 + ((149-140)/20)*(Tsdome-7))*10^(-7);
NuDomeforced = 2 + (0.4*(ReDomeforced^0.5) + 0.06*(ReDomeforced^(2/3)))*(Pr^0.4)*((muTa/muS)^0.
hdomeforced = NuDomeforced*k/D;

% Net h

%Correlation for cylinders and spheres is used to find the net h for the 2
%bodies.
hcylinder = (hcyl^3.5 + hcylforced^3.5)^(1/3.5);
hdome = (hdome^3.5 + hdomeforced^3.5)^(1/3.5);


% used qconduction + qradiation (solar radiation will always add heat to
% the system) = qconvection
% qconduction = (thermal conductivity)*width*(Tinner - Tsurface)
% qradiation is found using Omkar's analysis. The value taken is 160 W/m2
% qconvection = (HT coeffecient)*(Surface area)*(Tsurface - Tambient)

Tsactualcylinder = (kwall*w*Ti + hcylinder*pi*D*L*Ta + 160*pi*D*L)/(pi*D*L*hcylinder + kwall*w)
Tsactualdome = (kwall*w*Ti + hdome*(pi*D*D/2)*Ta + 160*(pi*D*D/2))/((pi*D*D/2)*hdome + kwall*w)
end

Tsactualcylinder %Final outerwall temperature of the cylinder
```

```
Tsactualcylinder = -47.6756
```

```
Tsactualdome %Final outerwall temperature of the dome
```

```
Tsactualdome = -34.1019
```

```
qlosscylinder = hcylinder*(Tsactualcylinder-Ta)
```

```
qlosscylinder = 160.8871
```

```
qlossdome = hdome*(Tsactualdome - Ta)
```

```
qlossdome = 160.8572
```

```
qloss = qlossdome + qlosscylinder %total heat loss from both the bodies combined
```

```
qloss = 321.7443
```